MINING DEALER BEST PRACTICE

3516 ENGINE CRANKSHAFT COUNTERWEIGHT INSTALLATION/REMOVAL TOOL

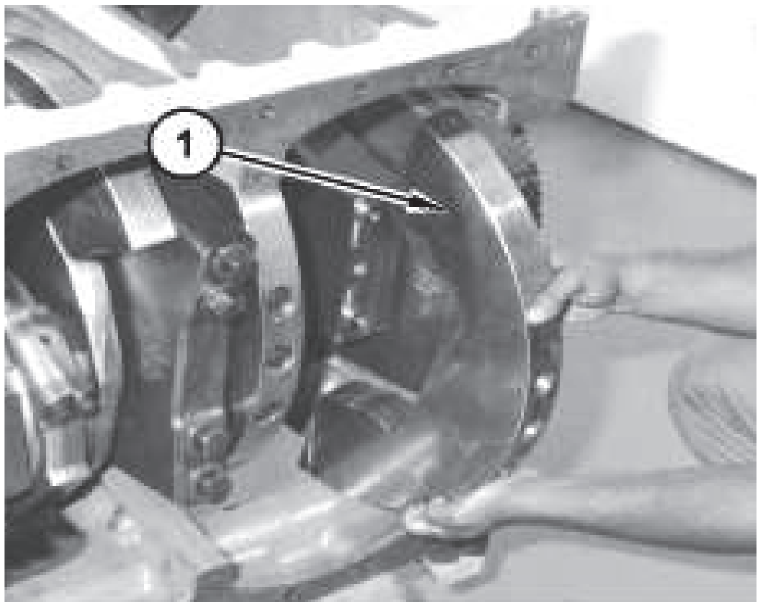
Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management

1.11-160714-03

1.0 Introduction

During Engine rebuilds, crankshafts must be visually inspected and the measurements should be checked to determine if they can be reused. The counterweights are part of the crankshaft and are critical for the balance and durability of the crankshaft. It is necessary to remove the counterweights to conduct a visual inspection to verify friction and signs of damage to the contact surfaces or in the holes where the bolts are installed.

The removal process of the 16 crankshaft counterweights is very repetitive and tedious because they are removed in groups of 4 counterweights, and every time a group is removed, the crankshaft must be turned to continue with the next group, this causes lumbar type problems in the workers if no mechanical aid is used.



Through a study conducted in the work area, it was determined that the personnel working on the counterweight removal were prone to musculoskeletal problems because it was a repetitive activity and it was suggested to manufacture a mechanical device that would facilitate such task. This best practice aims to design a lifting tool for handling the counterweight to minimize the aforementioned problems.

2.0 Best Practice Description

The tool consists of a steel trapezoid plate with an opening at the top where the lifting equipment is hooked as well as two rings welded on the bottom corners of the trapezium. Each ring is joined through a ¼" connector, a 1/8 – 3/16 x 0.5 m basket-shaped plasticized sling with thimbles at both ends. The approximate dimensions of the tool are: Depth 200 mm x Width 25.4 mm x Height 223 mm and the weight is approximately Weight: 2.15 Kg.

3.0 Implementation Steps

Step 1: Determine the size and weight of the counterweights the tool is to be used with. In this case work was being done on a 3516 engine. On the SISWEB, no tool is shown for this activity, despite the part weighs about 13 kg, because it is a repetitive activity, it creates lumbar type problems to the technicians.

Step 2: Design the lifting tool. Due to the limited space available between the counterweight and the crankshaft, it was necessary that the tool be versatile and easily accessible. The tool in the attached pictures and drawing was designed, a steel sling is placed in each ring taking into account the geometry of the counterweights.

Step 3: Build lifting tool. Initially, a plate that serves as the tool structure is made, a drilling is also made where the lifting equipment hook is entered and the two rings are welded at the bottom of it.

Step 3: The tests are performed and the appropriate length of the steel slings and the plate is checked.

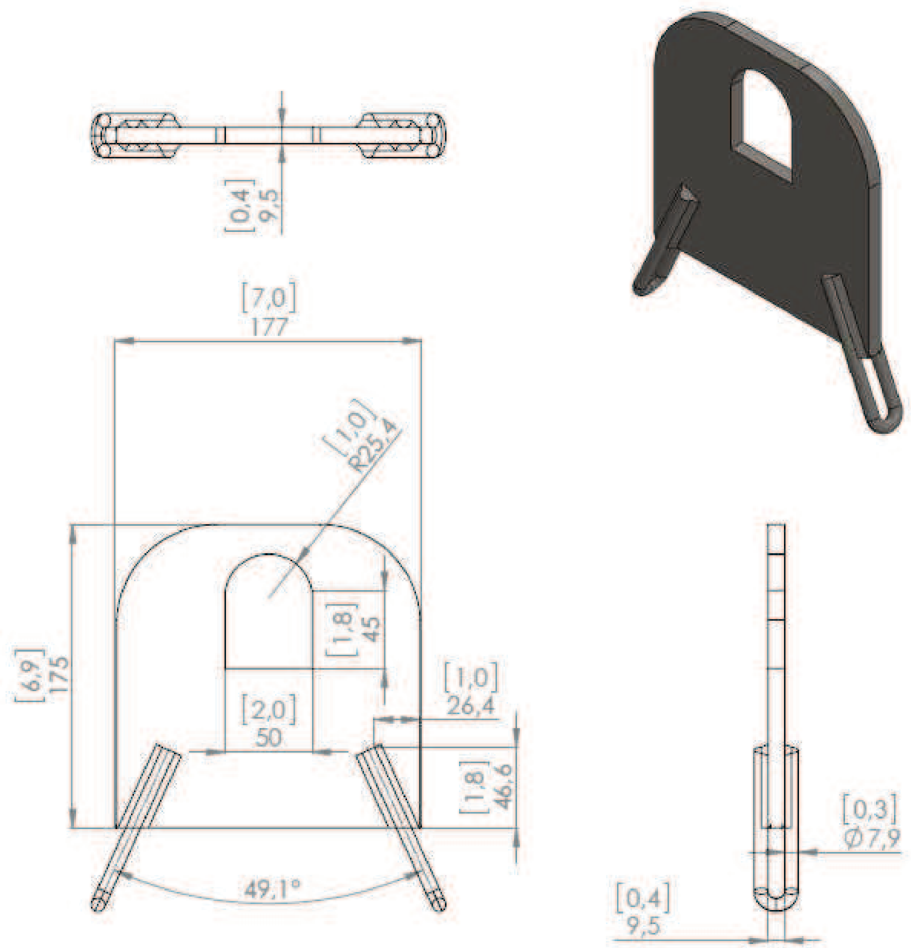
4.0 Benefits

Technician safety - Use of the lifting tool in this best practice reduces the risk of injury due to repetitive lifting by the technicians removing engine counterweights.

5.0 Resources Required

- Total Investment: Slings and Connectors: 211,000 COP plate and rings: 300,000 COP
- Crankshaft Counterweights lifting tool Memory calculation
- Equipment and Tools: Welding, Lifting accessories, steel sheet, steel cutting equipment

6.0 Supporting Attachments / References

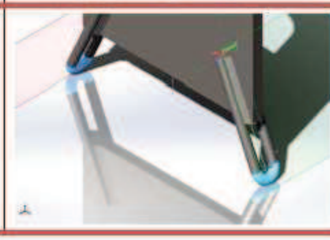


Validation efforts to load application:

Nombre de sujeción	Imagen de sujeción	Detalles de sujeción
Fijo-1		**Entidades:** 1 cara(s) **Tipo:** Geometría fija

Fuerzas resultantes

Componentes	X	Y	Z	Resultante
Fuerza de reacción(N)	-0.0212046	-13.5542	0.0114524	13.5542
Momento de reacción(N.m)	0	0	0	0

Nombre de carga	Cargar imagen	Detalles de carga
Gravedad-1		**Referencia:** Planta **Valores:** 0 0 -9.81 **Unidades:** SI
Fuerza-1		**Entidades:** 2 cara(s) **Tipo:** Aplicar fuerza normal **Valor:** 130 N

Nombre	Tipo	Mín.	Máx.
Factor de seguridad1	Automático	776.531 Nodo: 35693	7.13866e+006 Nodo: 15271

3 Izaje Contrapesa Cigueñales-Análisis estático 1-Factor de seguridad-Factor de seguridad1

Safety factor greater than 5 complying the lifting tools standards.

7.0 Related Best Practices

None

8.0 Acknowledgements

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